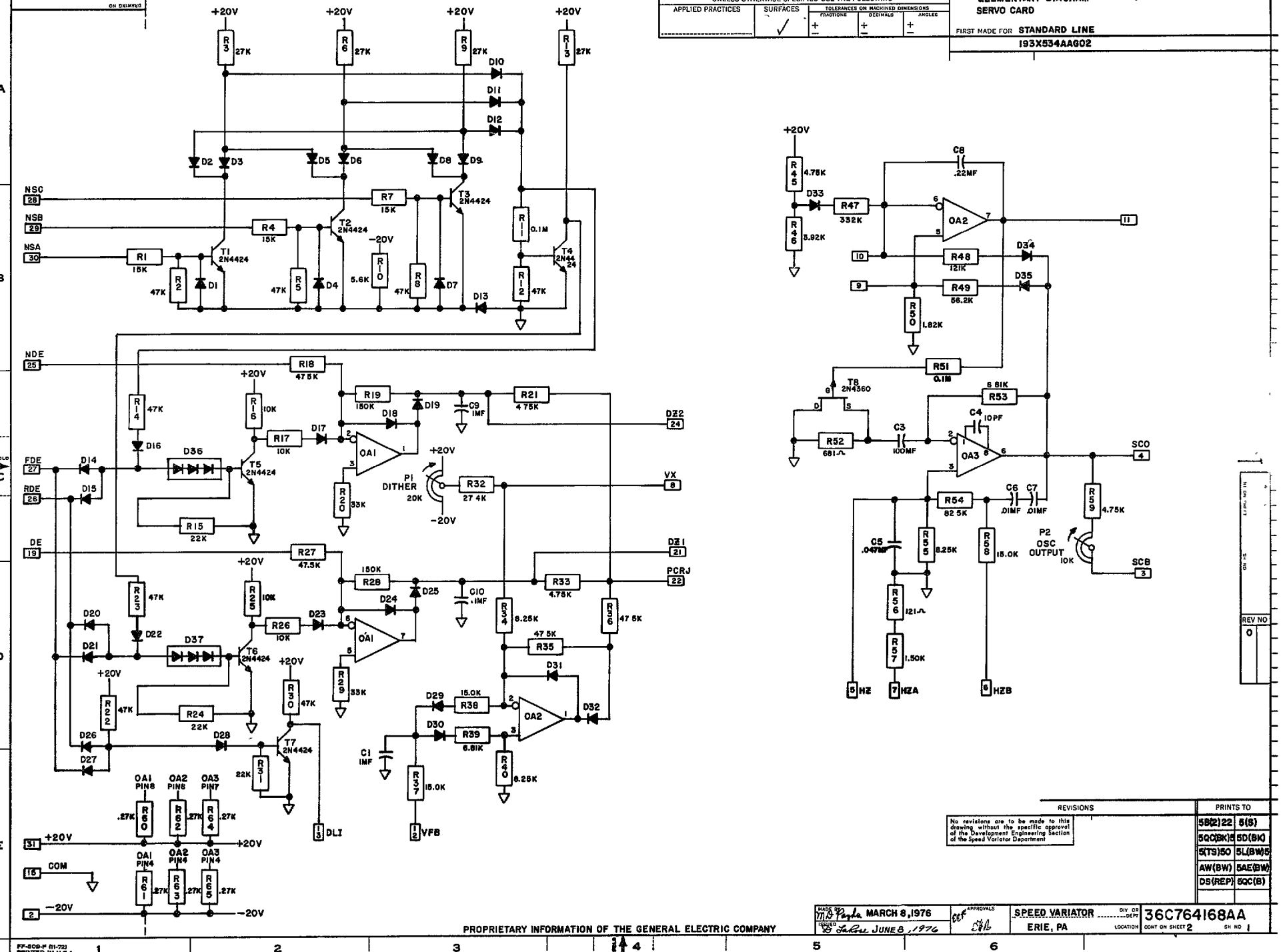
SERVO
193X534AA G02

ELEMENTARY DIAGRAM	36C764168AA SH. 1
CARD DIAGRAM	36C764168AA SH. 2

REFER TO THE DRIVE SYSTEM INSTRUCTIONS AND ELEMENTARY DIAGRAMS FOR DESCRIPTIONS, ADJUSTMENTS AND ON-CARD JUMPER CONNECTIONS.

These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

GENERAL ELECTRIC



GENERAL ELECTRIC

36C764168AA
CONT ON SHEET 2 SH NO 1

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING —
APPLIED PRACTICES | SURFACES ✓ | TOLERANCES ON MACHINED DIMENSIONS: FRACTIONS ± | DECIMALS ± | ANGLES ±

TITLE ELEMENTARY DIAGRAM
SERVO CARD

FIRST MADE FOR STANDARD LINE
193X534AA902

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

REV NO 0

PRINTS TO

5B(2)22	5(S)
5QC(BK)5	5D(BK)
5(T3)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

MADE BY M D Payla MARCH 8, 1976
ISSUED B LaRue JUNE 8, 1976
APPROVALS

SPEED VARIATOR DIV OR DEPT
ERIE, PA LOCATION

36C764168AA
CONT ON SHEET 2 SH NO 1

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

FF-502-F (11-72)
PRINTED IN U.S.A.

GENERAL ELECTRIC

36C764168AA
CONT ON SHEET FL SH NO 2

UNLESS OTHERWISE SPECIFIED USE THE FOLLOWING -

APPLIED PRACTICES	SURFACES	TOLERANCES ON MACHINED DIMENSIONS FRACTIONS	DECIMALS	ANGLES
710ASIS	✓	±	±	±

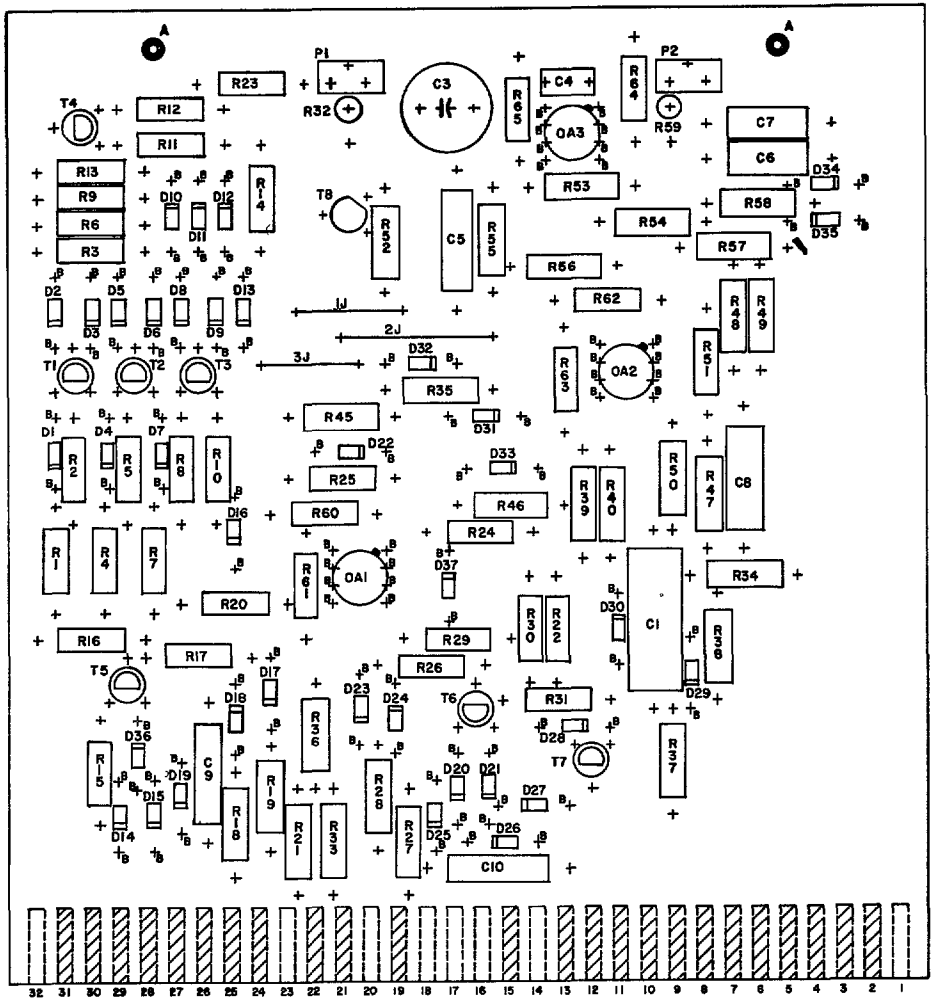
TITLE PRINTED CIRCUIT DIAGRAM
SERVO CARD

FIRST MADE FOR STANDARD LINE
193X534AAG02

HOLE TABULATION

ALL HOLES .040 DIA. EXCEPT THE HOLES TABULATED BELOW

LOC.	DIA.	QUAN.
A	.157	2
B	.032	98

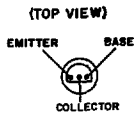


GROUP	KEY LOCATIONS
G02	8—9 12—13 28—29

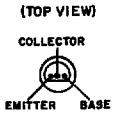
NOTES

1. INDICATED TAB NUMBERS CORRESPOND TO MATCHING RECEPTACLE NUMBERS
2. CROSS HATCHED TABS INDICATES TABS USED.
3. CARD SIZE, $5.500^{+.000}_{-.015}$ X $5.130^{+.002}_{-.008}$

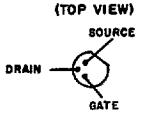
TRANSISTOR LEAD SKETCHES



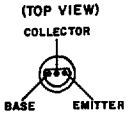
SKETCH FOR T7



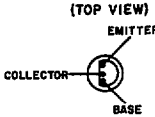
SKETCH FOR T1, T2, T3, T5



SKETCH FOR T8



SKETCH FOR T6

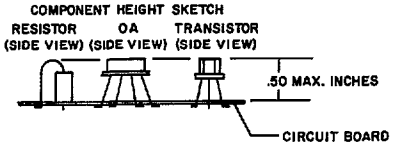


SKETCH FOR T4



SKETCH FOR OA1, OA2, OA3

COMPONENT HEIGHT SKETCH



2X SCALE	36A358024AA004 REAR ETCHED BOARD DWG.	FRONT ETCHED BOARD DWG.	1 FRONT SPACE UNIT REQUIREMENT	BACK

REVISIONS

No revisions are to be made to this drawing without the specific approval of the Development Engineering Section of the Speed Variator Department

PRINTS TO	
58(2)22	5(S)
5QC(BK)5	5D(BK)
5(TS)50	5L(BW)5
AW(BW)	5AE(BW)
DS(REP)	5QC(B)

REV NO
D

PROPRIETARY INFORMATION OF THE GENERAL ELECTRIC COMPANY

MADE MD Pugh MARCH 9, 1976	APPROVALS	SPEED VARIATOR	DIV OR DEPT	36C764168AA
ISSUED L.R. Rau JUNE 8, 1976		ERIE, PA	LOCATION	CONT ON SHEET FL SH NO 2

PROCESS CONTROL FOLLOWER

Printed Circuit Card PCF (Process Control Follower) provides circuitry to convert the current signal of a process controller to a voltage signal. This voltage signal provides a speed reference for the adjustable speed drive. Card PCF is mounted in the space to the right of relay RUN and above 2TB. In some cases, circuitry like that of PCF may be packaged on a component board in the modification rack, MDR. If so, these instructions will still generally apply.

The process controller is connected between terminals, PCP and PCN. The positive controller output is connected to PCP.

Jumpers on PCF are used to select resistors that convert various controller current ranges to the reference voltage output at PCF (ASR).

Jumper connections are:

Controller Current	0–5ma	1–5ma	4–20ma	10–50ma
PCF Jumpers	none	none	P to 20	P to 50

MCC: Remove jumper between SRH and COM for a 5 to 9V maximum input reference.

If an alternate reference mode is provided, set MCC's REF SCALE, MAX SPEED and MIN SPEED (If provided) potentiometers for the alternate reference as described in the System Instruction Book. (The alternate reference must be in the range of 5 to 9V maximum).

With minimum process controller output, set PCF's MIN potentiometer for zero (or minimum) reference (or speed).

With maximum controller output, set the MAX (PCF) potentiometer for maximum speed (approximately –10 volts at test post MCC–TR).

If no alternate reference is provided, set MAX (PCF) and REF SCALE (MCC) at mid range, and set MAX SPEED (MCC) full CCW. Set REF SCALE such that MCC–TR is –10 volts with the controller at maximum output. Set MAX SPEED for correct speed. Adjust MIN for zero (or minimum) speed with minimum controller output. Adjust MAX for correct speed at maximum controller output.

Card PCF also contains a JOG potentiometer which may be used for setting speed when the jog option is provided